5th place solution with code

Rank	5
Team	more CV challenge pls
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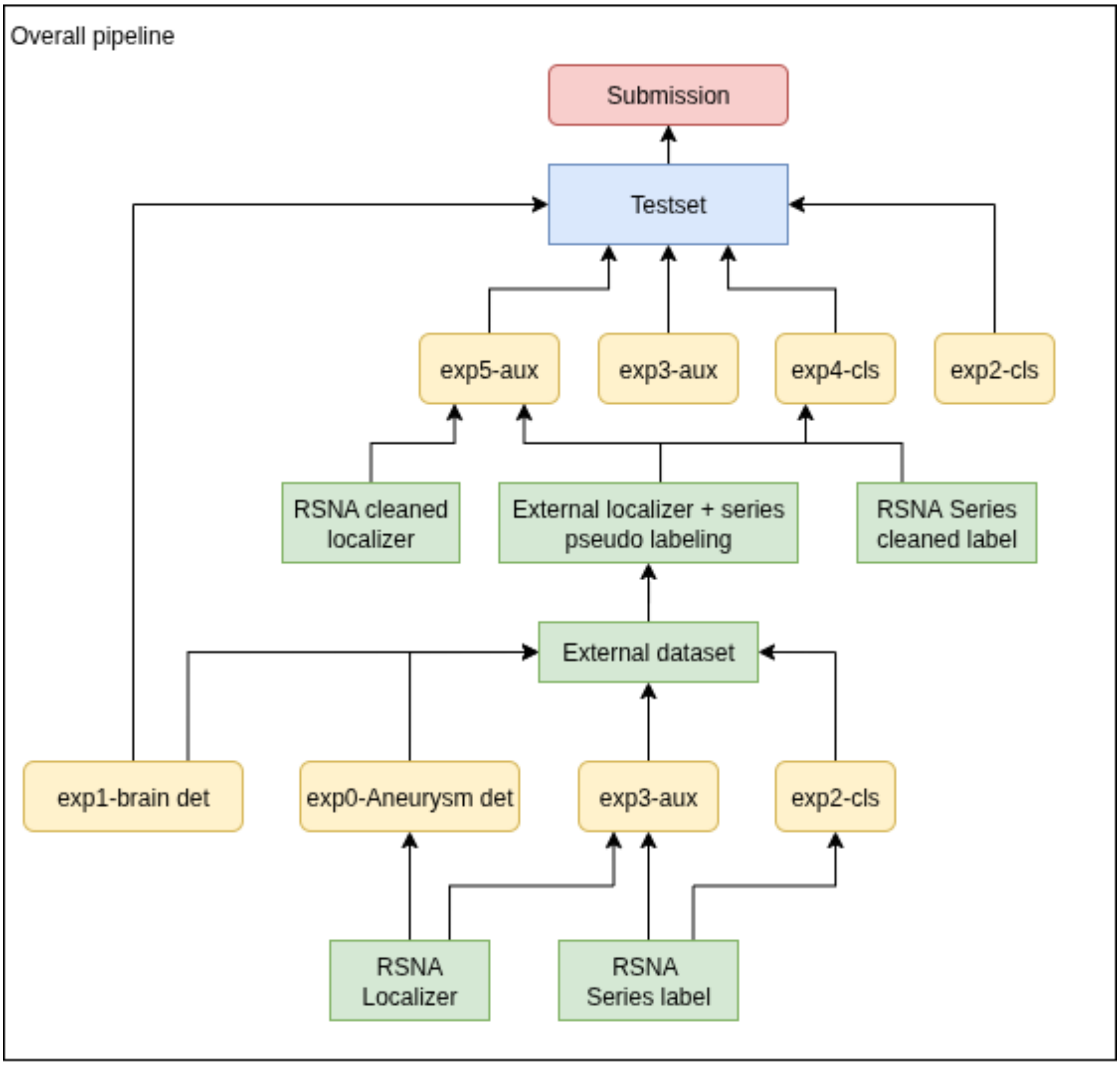
Topic ID	611849
Version	Original

Source: https://www.kaggle.com/competitions/rsna-intracranial-aneurysm-detection/discussion/611849
Retrieved with Kaggle CLI on 2026-07-07.

Thanks to Kaggle and RSNA for hosting this exciting competition.
It was a great learning experience and it was very interesting to see how much of my computer vision experience could also be applied to medical imaging. I'm quite disappointed with the result, but I see it as an opportunity to learn from other teams.

Github code: https://github.com/hoanghuyen797/RSNA-Intracranial-Aneurysm-Detection
Inference notebook: https://www.kaggle.com/code/longb173/rsna-iad-final-nb?scriptVersionId=266774552
Demo notebook: https://github.com/hoanghuyen797/RSNA-Intracranial-Aneurysm-Detection/blob/main/src/demo-test/test.ipynb

Overall Pipeline

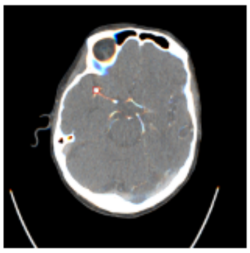
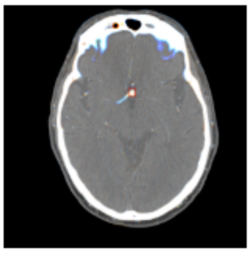
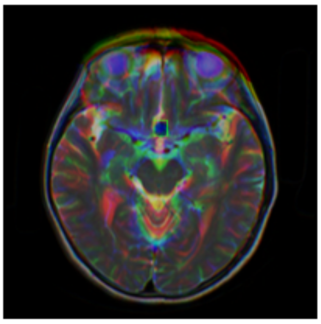
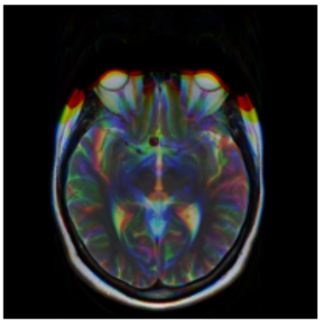


1. 2.5D image

For each slice t, I combine slices t-1 and t+1 to create a 3-channel image corresponding to [t-1, t, t+1]

2. Exp0: Aneurysm detection

In my experience, using only the classification labels (train.csv) is not as accurate as combining the classification and localization labels (train_localizers.csv). Therefore, based on the labels provided in train_localizers.csv by the host, for each aneurysm centroid, I searched within ±10 neighboring slices and manually annotate bounding boxes for the aneurysm using LabelImg.
This process does not require specialized medical knowledge since the aneurysm centroids are already provided.
2 classes: aneurysm (modality CTA, MRA, MRI T1post) and aneurysm_mri_t2 (modality MRI T2)

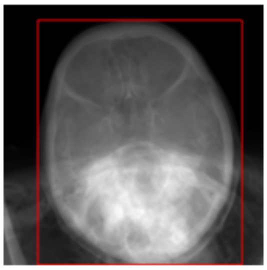
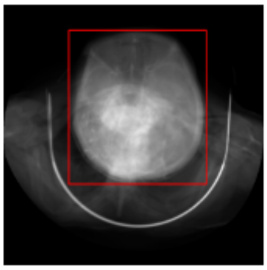
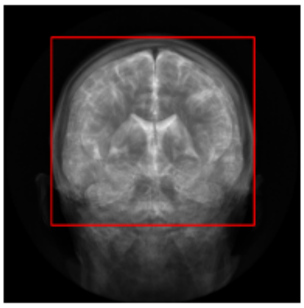
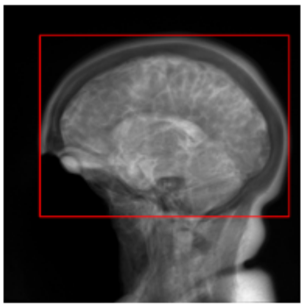
aneurysm	aneurysm	aneurysm_mri_t2	aneurysm_mri_t2
			

Then I train 5 models (5 folds) using YOLOv11x 1280

metric	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
mAP50	0.705	0.647	0.766	0.702	0.691
mAP50-95	0.460	0.429	0.504	0.482	0.449

3. Exp1: Brain detection

For each SeriesInstanceUID in the training set, I generate a single image by averaging all slices.
Then, I manually annotate the brain bounding box as the following 2 classes: brain (brain in axial view) and abnormal (brain in other views). Each slice in the series will be cropped according to the bounding box predicted by this model. This reduces background noise (especially for slices containing lung regions …), which improves the model's accuracy by about **0.03-0.05**.

brain	brain	abnormal	abnormal
			

Then I train 1 model using yolov5n 640: mAP50-95 = 0.948

4. Image augmentation

`train_transform = albu.Compose([
albu.RandomResizedCrop(size=(self.image_size, self.image_size), scale=(0.5, 1.0), ratio=(0.75, 1.3333), p=1),
albu.ShiftScaleRotate(rotate_limit=15, border_mode=0, p=0.5),
albu.OneOf([
albu.MotionBlur(blur_limit=5),
albu.MedianBlur(blur_limit=5),
albu.GaussianBlur(blur_limit=5),
albu.GaussNoise(var_limit=(5.0, 30.0)),
], p=0.5),
albu.CLAHE(clip_limit=4.0, p=0.5),
albu.HueSaturationValue(p=0.5),
albu.RandomBrightnessContrast(p=0.5),

val_transform = albu.Compose([
albu.Resize(self.image_size, self.image_size),
])`

Horizontal Flip: It may sound unreasonable, but I applied horizontal flipping to the images and adjusted the labels as following:
Left Infraclinoid Internal Carotid Artery <-> Right Infraclinoid Internal Carotid Artery
Left Supraclinoid Internal Carotid Artery <->Right Supraclinoid Internal Carotid Artery
Left Middle Cerebral Artery <->Right Middle Cerebral Artery
Left Anterior Cerebral Artery <->Right Anterior Cerebral Artery
Left Posterior Communicating Artery <->Right Posterior Communicating Artery
This worked, and my model's accuracy improved by about 0.01

5. Exp2: 2 classification models

2 multi-label classification models (14 classes), image size=384, trained on this competition dataset. For negative series I use all slices, while for positive series I only use slices that contain aneurysm boxes (as

reviewed in section 2). The label of each slice is the same as the series label in the train.csv
vit large 384: OOF AUC = 0.8503
eva large 384: OOF AUC = 0.8551
Due to time constraints, I couldn't use 5 models (5 folds) for prediction. I could only use a single model trained on almost the full dataset, using only 50 series for evaluation, and selecting the best epoch based on the public leaderboard.

6. Exp3: Multi-task classification + segmentation

1 model (mit b4 fpn), image size = 384, trained on the RSNA dataset. Similar to exp2, for negative series I use all slices, while for positive series I only use slices that contain aneurysm boxes (as reviewed in section 2). The label of each slice is the same as the series label in the train.csv. For the submission, I only used the prediction from the classification task. To create masks for the segmentation task, I use the aneurysm bounding box (2).
mit-b4 FPN : OOF AUC = 0.8549

7. External dataset

I use the competition dataset and 2 external datasets:
Lausanne_TOFMRA: https://openneuro.org/datasets/ds003949/versions/1.0.1
Royal_Brisbane_TOFMRA: https://openneuro.org/datasets/ds005096/versions/1.0.3
For the external data, I generate the series label using prediction of 2 models (vit large exp2 + mit b4 exp3), and generate the localization label using prediction of model exp0-aneurysm detection

8. Clean trainset

For negative series in the trainset with 'Aneurysm Present' prediction score of 2 models (vit large exp2 + mit b4 exp3) > 0.9, I will change it to positive and use model exp0-aneurysm detection to create localization label

9. Exp4: 2 classification models

2 classification models trained on the cleaned RSNA dataset (8) and external dataset (Lausanne_TOFMRA + Royal_Brisbane_TOFMRA) pseudo labeling (7)
vit large 384: OOF AUC = 0.8558
eva large 384: OOF AUC = 0.8579

10: Exp5: Multi-task classification + segmentation

1 model (mit b4 fpn), image size = 384, trained on the cleaned RSNA dataset (8) and external dataset (Lausanne_TOFMRA + Royal_Brisbane_TOFMRA) pseudo labeling (7)
mit-b4 FPN 384: OOF AUC = 0.8629

11. Final submission

The final submission is the ensemble of 6 models:

- Final1: 0.25 exp3_mit_b4 + 0.25 exp5_mit_b4 + 0.125 exp2_vit_large + 0.125 exp4_vit_large + 0.125 exp2_eva_large + 0.125 exp2_eva_large
- Final2: 0.5 exp3_mit_b4 + 0.125 exp2_vit_large + 0.125 exp4_vit_large + 0.125 exp2_eva_large + 0.125 exp2_eva_large

Final1: OOF AUC = 0.8823, public LB = 0.89
Final2: OOF AUC = 0.8767, public LB = 0.89

Supplemental Q&A

> **LSL000UD · 2025-10-15T06:16:07.807000**
>
> Congratulations on your great results, nice work! Regarding using the model to clean the dataset, how many cases did you clean in total?

> **HoangHuyen · 2025-10-15T06:29:03.127000**
>
> 33 negative cases

> **Satwik · 2025-10-15T06:31:02.227000**
>
> Congratulations! This is a great solution , 2.5D seems to work wonders in this competition. I have a few questions (some may sound stupid, sorry) -
>
> 1. Is there a reason why you manually annotated bboxes, and also separated MRI T2 aneurysms as a separate class? I trained a similar yolo12L to detect aneurysms , searching in 7 neighboring slices using 1D images (each slice as image) and it could get mAP50 0.8+.
> 2. ~0.85 AUC using only classification models is really impressive (exp2). Are they trained using 2.5D slices only?
> 3. How do you do inference with the slice-wise classification models? Do you infer on all slices , and then aggregate predictions?
>
> Btw, the codebase is really concise, well maintained and easy to read. A lot to learn from you.
>
> Thank you, and congratulations again!

> **HoangHuyen · 2025-10-15T06:53:36.160000**
>
> 1. In MRI T2, the aneurysm seems dark, unlike in other modalities where it appears bright (based on the observed centroids).
> 2. Yes 2.5D, you can check it on my GitHub repo
> 3. Yes, and the series prediction is the maximum in z-axis (slice): num_slicesx14 -> 14 probs

> **Hide on bush · 2025-10-15T08:05:23.100000**
>
> I would like to ask how the final model ensemble weights were calculated?

HoangHuyen · 2025-10-15T08:22:06

try a few weight combinations to achieve the highest OOF AUC and the highest public LB